IAM and AWS Tools – Key Points

IAM

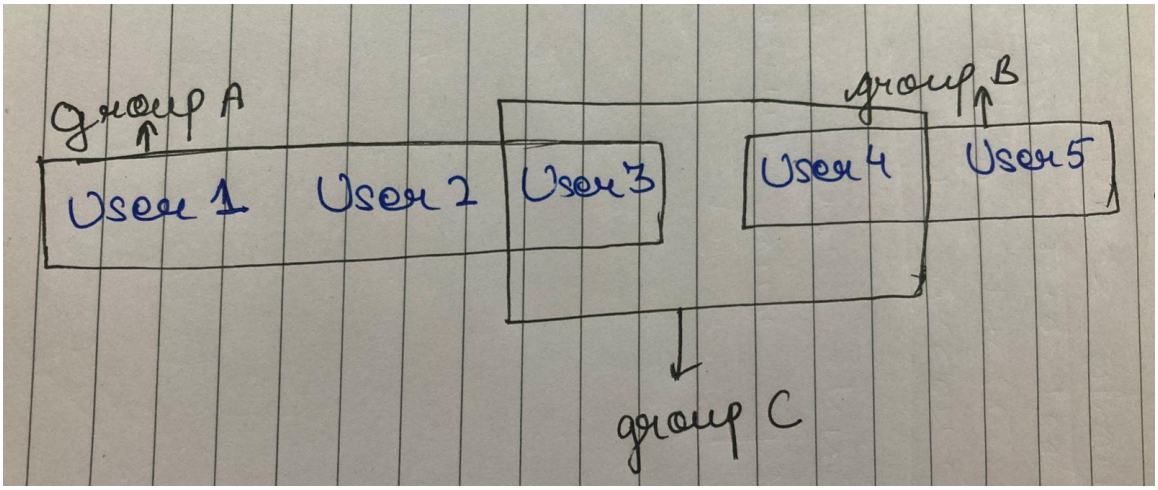
IAM (Identity and Access Management) is an AWS service that enables you to securely control access to AWS services and resources.

It allows you to create users, groups, and roles, and assign permissions using policies to define what each entity can access or perform.

IAM ensures fine-grained access control and helps enforce security best practices like least privilege and multi-factor authentication.

IAM Users, Groups, and Policies

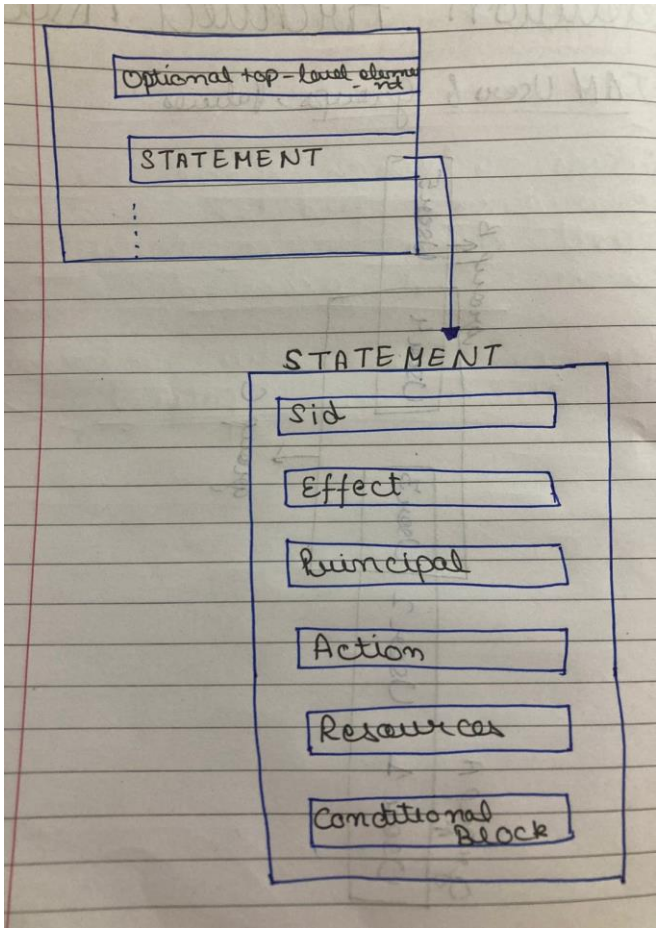
- Users: Individuals with long-term credentials to access AWS.
- Groups: Collection of users sharing the same permissions.
- Policies: JSON documents that define permissions. Attached to users, groups, or roles.
- One user can belong to multiple groups.



- Here, User 1, User 2, and User 3 belong to group A.
- User 4, and User 5 belong to group B.
- User 3 and User 4 belong to group C. Thus, one user can belong to multiple groups.

IAM Policies Structure

- Types: Managed (AWS or Customer), Inline, Identity-based, Resource-based.
- Structure: Version, Statement (Effect, Action, Resource, Condition).



- **Version**: Policy language version. Use "2012-10-17" (latest and recommended).
- **Statement**: Main container for individual permission rules (can be a list).
- **Effect**: Either "Allow" or "Deny" – defines whether to permit or block the action.
- **Action**: AWS service actions (e.g., s3:PutObject, ec2:StartInstances).
- **Resource**: ARN of the resource the action applies to (e.g., S3 bucket, EC2 instance).
- Condition (optional): Additional constraints (e.g., IP address, MFA required, time of day).

AWS CLI

AWS CLI (Command Line Interface) is a tool that lets you **interact with AWS services using commands** in your terminal or command prompt.

It enables you to **manage resources** (like S3, EC2, IAM, etc.) by running commands instead of using the AWS Management Console.

The CLI is ideal for **automation, scripting**, and **quick operations**, and supports features like profiles, output formatting, and credential management.

Access Keys, CLI, and SDK

- Access Keys: Consist of Access Key ID and Secret Access Key. Used for programmatic access.
- AWS CLI: Command Line Interface to interact with AWS services.
- AWS SDK: Language-specific tools (Python - boto3, JavaScript - aws-sdk, etc.) to access AWS programmatically.

AWS CloudShell

- Browser-based shell with pre-installed AWS CLI and tools.
- No setup needed, runs in AWS environment with IAM permissions.

IAM Roles and Services

- Roles: Provide temporary credentials to trusted entities (users, apps, services).
- Used by EC2, Lambda, and other services to access AWS resources securely.

IAM Security Tools

- IAM Access Analyzer: Identifies resources shared with external entities.
- Credential Report: Lists all users and the status of their credentials.
- IAM Policy Simulator: Test and troubleshoot policies.